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Amkor Technology

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Capital Expenditure Supercycle Dampens Advanced Packaging Margin Triumphs

Analyst: Justin Wen | October 29, 2026

Amkor Technology is demonstrating remarkable operational leverage across its core advanced packaging franchises, highlighted by record second-quarter fiscal 2026 revenue of $1.90 billion and gross margins reaching 16.8%. However, the market has excessively discounted the cash flow friction resulting from the company's aggressive $2.5-3.0 billion fiscal 2026 capital expenditure program dedicated to the Arizona campus and High-Density Fan-Out expansion. With first-half free cash flow falling to -$307 million alongside a multi-year absorption cycle for domestic packaging infrastructure, capital intensity will suppress near-term shareholder returns despite cyclical demand recovery in communications and automotive end markets. We initiate coverage with a Hold rating, establishing a probability-weighted blended target of $37.14, representing -10.5% downside from the reference spot price of $41.50.

HOLD	$37.14	-10.5%	$41.50
INVESTMENT RATING	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE

The broader market remains fixated on near-term gross margin acceleration and triple-digit net income expansion fueled by artificial intelligence packaging demand. This optimism overlooks the structural drag of sustained negative free cash flow and looming second-half equipment depreciation associated with the $2.5-3.0 billion

Arizona buildout. Multi-year cash generation will remain constrained until commercial utilization on domestic high-density lines scales to parity with overseas facilities.

Executive summary

Amkor Technology operates as an outsourced chip assembly and test (OSAT) company that packages semiconductors for major chipmakers across the globe. The company's competitive footprint centers on complex assembly platforms, primarily advanced packaging solutions that encompass High-Density Fan-Out (HDFO) and 2.5D architectures, alongside conventional packaging capabilities. Amkor serves a diversified customer base across four critical end markets, consisting of communications, automotive and industrial, computing, and consumer electronics. In recent fiscal quarters, the business has pivoted decisively toward advanced products, which accounted for $1.557 billion or 82% of total sales in the second quarter of fiscal 2026, driven by high-performance computing, artificial intelligence workloads, and premium smartphone launches.

The company's recent operational trajectory reflects a dramatic cyclical recovery coupled with significant operational leverage. After posting fiscal 2025 revenue of $6.71 billion with an operating margin of 7.0%, performance inflected positively throughout early fiscal 2026. First-quarter fiscal 2026 net sales expanded 27% year over year to $1.685 billion as operating margin rose to 6.0% from 2.4% in the prior year period, supported by the Vietnam facility moving past breakeven. This acceleration continued into the second quarter of fiscal 2026, where Amkor delivered record revenue of $1.898 billion, representing 26% year-over-year growth. Gross margin widened by 480 basis points year over year to 16.8%, and net income tripled to $174 million, translating to diluted earnings per share of $0.70. Management guidance for the third quarter of fiscal 2026 points toward approximately $2.0 billion in revenue with gross margin breaking 19%.

Notwithstanding this operational strength, capital allocation and free cash flow generation face acute structural challenges. Amkor concluded fiscal 2025 with $1.38 billion in cash and total debt of $1.56 billion, yielding net debt of $182.6M. However, management initiated a massive capital deployment phase, budgeting $2.5-3.0 billion in fiscal 2026 capital expenditures, nearly tripling the $905 million deployed in fiscal 2025. This expansion centers on the Arizona campus, where total planned investment was raised to $7 billion, alongside worldwide HDFO capacity additions. Consequently, first-quarter fiscal 2026 free cash flow registered at -$79 million, followed by an H1 fiscal 2026 free cash flow deficit of -$307 million on $688 million of capital outlays. With full-year capex representing more than 35% of annualized revenue, the sheer scale of investment creates an acute medium-term cash flow drag that counterbalances volume leverage.

Operational snapshot

$41.50	$6.71B	$191.0M	2.8%	1.30	$41.50
REF. PRICE	FY2025 REVENUE	FY2025 FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

Fiscal-year figures as reported in company filings and collected by Cascadia Analytics; ref. price as of 2026-10-29.

The moat is Capital-Intensive and Technologically Entrenched

Amkor's competitive advantage in outsourced semiconductor packaging and testing stems from immense engineering complexity, long qualification cycles, and deeply intertwined customer integrations. As an established OSAT partner to leading global chipmakers, Amkor is structurally embedded in the design and manufacturing roadmaps of premium mobile processors, high-performance computing accelerators, and mission-critical automotive subsystems. Transferring high-volume, high-density packaging programs to alternative vendors requires extensive re-qualification, tooling reinvestment, and yield stabilization periods that semiconductor designers avoid during active product life cycles. The technical expertise required to successfully manufacture multi-die packages, wafer-level fan-out systems, and 2.5D assemblies forms a formidable operational barrier against lower-tier competitors.

The depth of this entrenchment is evident in the execution of key product ramps. In the third quarter of fiscal 2025, revenue surged 31% sequentially to $1.987 billion on a 67% sequential expansion in communications revenue, directly tied to an iOS premium smartphone ramp. Such precise seasonal execution highlights Amkor's status as a primary manufacturing partner for the world's most demanding consumer and mobile hardware providers. Furthermore, the operational turnaround of overseas sites reinforces this technical scale, as demonstrated in the first quarter of fiscal 2026 when the company's Vietnam manufacturing facility moved past breakeven, adding scalable, cost-effective capacity to handle high-volume advanced product lines.

In parallel, Amkor is solidifying its strategic relevance through geographic diversification and onshore advanced packaging capacity. By raising its planned Arizona campus investment to $7 billion, the company is positioning itself as the foundational cornerstone for the domestic United States semiconductor supply chain. This multi-year initiative directly addresses chipmakers' mandates for localized advanced packaging ecosystems that align with domestic front-end wafer fabrication. This capital commitment cements customer retention, yet it simultaneously establishes an operational hurdle that requires consistent high-margin volume to absorb significant fixed overhead.

Customer pricing power remains asymmetric across Amkor's portfolio. While proprietary advanced packaging architectures command premium pricing and volume commitments, traditional packaging solutions remain subject to broader semiconductor cycle fluctuations. This dichotomy was visible in the consumer end market, which contracted 10% year over year in the

fourth quarter of fiscal 2025, fell 5% in the third quarter of fiscal 2025, and dropped 2.5% in the second quarter of fiscal 2026, representing the only shrinking end market across the company's portfolio. Amkor's moat is therefore reinforced when capital is allocated toward complex advanced platforms, while exposure to commoditized, price-sensitive legacy consumer nodes dampens aggregate returns.

Valuation

Three scenarios make the valuation range explicit

Our valuation framework evaluates Amkor's intermediate-term cash flow generation across the fiscal 2026 to fiscal 2030 forecast window. The intrinsic valuation relies on a rigorous discounted cash flow model calibrated against an 8.64% weighted average cost of capital, an equity beta of 1.30, and a long-term base terminal growth rate of 2.0%. Diluted share count is modeled at 250.0 million shares against a market capitalization of $10.4 billion at the reference spot price of $41.50.

The valuation range highlights the operational sensitivity between capital project execution and free cash flow conversion. Amkor generated $191.0 million in free cash flow during fiscal 2025, delivering an FCF margin of 2.8% on $6.71 billion of revenue. The baseline model assumes top-line revenue expands at an 8.3% compound annual growth rate through fiscal 2030, with target free cash flow margins normalizing to 5.5% as the $2.5-3.0 billion capex cycle matures and the Arizona and HDFO lines reach commercial density. In contrast, downside conditions reflect prolonged cash burn and sub-par asset utilization, while the upside scenario captures unconstrained AI packaging monetization and rapid domestic factory absorption.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	4.0%	3.9%	$15.31	-63.1%
Base	50%	8.3%	5.5%	$34.57	-16.7%
Bull	25%	12.7%	7.0%	$64.10	+54.5%

Bear $15.31

Severe front-loaded depreciation headwinds combine with ongoing consumer weakness and underutilization at the Arizona campus. Revenue compounds at a subdued 4.0% CAGR through fiscal 2030 while target FCF

Base $34.57

Balanced adoption of advanced packaging platforms supports an 8.3% revenue CAGR through fiscal 2030. Free cash flow margins normalize to 5.5% as capital outlays moderate from peak levels, producing

Bull $64.10

High-Density Fan-Out and 2.5D packaging platforms accelerate alongside robust automotive and communications demand, yielding a 12.7% revenue CAGR. Target FCF margin reaches 7.0%, generating an equity valuation of

margin stalls at 3.9%, driving a -63.1% decline relative to spot.

an intrinsic base value of $34.57, or -16.7% vs spot.

$64.10, representing +54.5% upside.

Valuation framework

A live discount rate, not a static hurdle

We discount Amkor's projected unlevered free cash flows using an 8.64% weighted average cost of capital. This rate incorporates an equity beta input of 1.30, capturing the inherent cyclicality of outsourced semiconductor assembly and test services, the operating leverage embedded in high fixed-cost manufacturing facilities, and the financial risk associated with elevated capital investment cycles. The discount rate reflects appropriate risk compensation for an OSAT participant navigating multi-billion-dollar domestic factory expansion while supporting global customer production nodes.

At the current reference spot price of $41.50, the market implies a cost of capital or cash flow trajectory that understates the drag of near-term capital intensity. With total debt standing at $1.56 billion against $1.38 billion in cash reserves, Amkor's net debt position of $182.6 million appears manageable. However, the requirement to fund a $2.5-3.0 billion fiscal 2026 capex envelope creates intermediate liquidity obligations that raise the operational risk profile, justifying our disciplined 8.64% hurdle rate across all forecast periods.

FCF margin normalization is the engine

The progression of Amkor's free cash flow margin serves as the primary operational swing factor across our three scenarios. In fiscal 2025, the company delivered a 2.8% FCF margin, generating $191.0 million in free cash flow from $6.71 billion in net sales. During the initial stages of the Arizona and HDFO investment cycles in the first half of fiscal 2026, free cash flow deteriorated sharply to -$307 million on capital expenditures of $688 million. This dynamic proves that near-term cash conversion will be severely suppressed by physical footprint expansion.

Our long-term model projects cash flow recovery as capital additions transition from active construction into high-volume manufacturing. In the base case, free cash flow margin expands to 5.5% by fiscal 2030 on an 8.3% revenue CAGR,

reflecting fixed-cost absorption across advanced packaging lines. In the bull case, where 2.5D and HDFO demand surges, target FCF margin expands to 7.0% alongside a 12.7% revenue CAGR. In the bear case, persistent capex commitments and delayed yield maturity restrict final-year FCF margin to 3.9% on a 4.0% revenue CAGR, resulting in severe valuation compression.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

> **Methodology.** Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

KEY VALUATION PARAMETERS

Forecast window	**FY2026 to FY2030**
Live WACC	**8.64% (2026-10-29)**
Terminal growth	**2.0%**
Equity beta	**1.30**
Diluted shares	**250.0 million**

HOLD

over $37.14 blended target

Competitive moat and concentration risk

We rate the moat as Narrow and we view the market structure as Oligopoly (OSAT). We assign a premium of +5% to sector median multiples on moat strength alone.

- scale and geographic footprint, $6.7bn 2025 revenue, 55+ years, advanced products >80% of Q4 2025 revenue, and a $2bn Arizona facility due 2028 (company/press coverage, 2025-2026).
- Gross margin expanded to 16.7% in Q4 2025 (reported Feb 2026) alongside a $30m strategic asset sale, margin expansion came with a one-off, not a price increase. No evidence of price-led margin expansion was found.
- The weakest element is no pricing power, Q4 2025 gross margin 16.7% and FY2025 net margin 5.6% (net income $373.9m on $6.7bn revenue), with top-10 customers at 72% of sales (Apple 29.8%, Qualcomm 11.1%). A 16.7% gross margin is a cost-competitor's margin, not a moat-pricing margin.

On a multiple cross-check, the justified range is 21.0-23.0x / 18.3-20.0x. The cross-check value is $27.63-$30.20, on a base of $1.51 median GAAP EPS, 2021-2025 (= FY2025 trailing). The gap verdict is COLLAPSES, not at all (below Base $34.42).

This figure uses the normalized base from the October 27 recheck, not the original boom-year base.

We have no disclosed backlog figure for this company, so backlog gives no support to the growth path here.

We score concentration and credit risk across seven families, where a High score is 8 points, a Medium score is 4 points, and a Low score is 1 point.

Risk family	Score
customer concentration	8 points
counterparty credit and funding dependence	1 points
single product, single program, or single platform dependency	4 points
single supplier, single fab, or single geography exposure	4 points
platform, channel, or distribution dependency	1 points
customer insourcing risk	8 points
regulatory single point risk	1 points

The combined discount is 20 percent, capped at 20 percent, against an uncapped total of 27 percent. Shown side by side with the moat premium or discount, the net stance is -15 pp points arithmetically. We do not net these figures into a new target, and we do not double count a family excluded under the double-count rule.

The ladder ranks risk coarsely, and because 32 of the 47 names reach the 20 percent cap, the combined discount is not a mechanical valuation haircut until the ladder is recalibrated.

On the gap to spot, the concentration assessment reads as follows. Not at all

Both the moat premium or discount and the concentration discount are multiple and risk cross-checks dated October 27, 2026, and neither figure is in the DCF target itself. Cross-check figures were computed against the Base in the October 27 framework, so they may differ from a live Base that has since moved.

Business & segment drivers

Revenue prospects anchor on Advanced Packaging Scale and Automotive Industrial Recovery HOLD

Amkor's top-line growth is anchored by its advanced products segment, which has become the primary operational engine of the enterprise. In the second quarter of fiscal 2026, advanced packaging revenue reached $1.557 billion, representing 82% of total corporate sales and growing 27% year over year. This followed a strong first quarter of fiscal 2026, where advanced products generated $1.37 billion, marking a 29% year-over-year increase and comprising 81% of total sales. The multi-year pivot toward advanced architectures is driven by surging demand for High-Density Fan-Out and 2.5D platforms, where revenue is projected to nearly triple throughout 2026 to support high-performance computing and artificial intelligence accelerators.

The communications end market remains Amkor's largest single exposure, subject to sharp seasonal inflections and customer smartphone refresh cycles. During the third quarter of fiscal 2025, communications revenue surged 67% sequentially on the back of an iOS premium smartphone ramp, pushing total corporate revenue to $1.987 billion. Momentum in communications persisted into fiscal 2026, advancing 40% year over year in the first quarter and approximately 32% year over year in the second quarter. The adoption of advanced packaging within premium mobile chipsets provides baseline volume, though it exposes gross margins to customer timing and packaging mix shifts.

Automotive and industrial end markets have confirmed an exit from their cyclical trough, delivering strong multi-quarter revenue recovery. In the fourth quarter of fiscal 2025, automotive and industrial revenue inflected positively, rising 18% sequentially and 22% year over year. By the second quarter of fiscal 2026, automotive and industrial sales surged approximately 38% year over year, reaching record quarterly revenue alongside computing. Computing has similarly benefited from accelerated data center demand, expanding 23% year over year in the third quarter of fiscal 2025 on artificial intelligence and high-performance computing requirements.

Conversely, consumer electronics continues to function as an operational drag across the enterprise. Consumer revenue contracted 5% year over year in the third quarter of fiscal 2025, fell 10% year over year in the fourth quarter of fiscal 2025, and declined another 2.5% year over year in the second quarter of fiscal 2026. As the sole contracting end market across recent quarters, consumer weakness partially offsets the gains achieved in communications and automotive packaging, illustrating why disciplined capacity allocation remains vital to defending overall operating margins.

What to monitor

- **Arizona campus construction progress and capital trajectory** Tracking whether fiscal 2026 capital expenditures exceed the projected $2.5-3.0 billion envelope as the $7 billion campus progresses.
- **Advanced packaging mix and HDFO adoption** Monitoring whether 2.5D and High-Density Fan-Out platform revenue meets expectations to nearly triple in 2026.
- **Depreciation drag on gross margin** Assessing the impact of front-loaded equipment purchases on second-half gross margin, following the 16.8% achieved in the second quarter of fiscal 2026.
- **Consumer market stabilization** Watching for an end to multi-quarter contractions after consumer revenue dropped 10% in the fourth quarter of fiscal 2025 and 2.5% in the second quarter of fiscal 2026.

+6.2%	14.0%	7.0%	2.8%	$182.6M	8.4%
FY2025 REVENUE GROWTH	GROSS MARGIN	OPERATING MARGIN	FCF MARGIN	NET DEBT	RETURN ON EQUITY

METRIC	FY2025 (AS REPORTED)
Revenue	$6.71B
Gross Profit	$938.6M
EBIT	$467.4M
Net Income	$373.9M
EBITDA	$1.16B
Free Cash Flow	$191.0M
Diluted EPS	$1.50
ROE	8.4%

Risk & moat matrix

The bear case

SHARE EROSION

Intense competition across core outsourced assembly programs can pressure baseline packaging volumes if competing OSATs add aggressive capacity. Shifts in customer multi-sourcing strategies across mobile platforms could undermine seasonal volume spikes in communications.

M&A DIGESTION

Large-scale physical expansions present operational execution challenges similar to complex asset integrations. Scaling greenfield infrastructure at the Arizona campus carries construction scheduling risks and delayed customer

REGULATORY AND PRICING SCRUTINY

Government packaging initiatives and subsidy scrutiny introduce structural compliance and operational reporting burdens. Shifting trade policies and regional assembly mandates can alter cross-border manufacturing flows between Asian facilities

TECHNOLOGY DISRUPTION

Rapid transitions in chiplet packaging architectures require continuous re-engineering of substrate and fan-out processes. Failure to keep pace with complex 2.5D and multi-die customer requirements could strand older assembly line investments.

REINVESTMENT TREADMILL

The fiscal 2026 capex envelope of $2.5-3.0 billion represents more than 35% of annualized revenue and nearly triples fiscal 2025 levels. Elevated capital requirements produced first-half fiscal 2026 free cash flow of -$307 million, creating substantial cash burn.

WACC AND DURATION SENSITIVITY

With an equity beta of 1.30 and an 8.64% discount rate, Amkor's valuation is vulnerable to persistent capital costs. Extended payback periods on greenfield factory investments depress the net present value of long-dated terminal cash flows.

qualification timelines.

and domestic sites.

The strengths

OLIGOPOLY POSITION

Amkor maintains a tier-one position as an outsourced semiconductor packaging and test provider for leading chipmakers worldwide. High technological hurdles in advanced packaging limit the number of vendors capable of executing complex high-volume programs.

FCF COMPOUNDING

Operating leverage inflected positively in the second quarter of fiscal 2026 as gross margin reached 16.8% and net income tripled to $174 million. Once peak factory investments subside, baseline cash generation can recover toward our 5.5% base FCF margin target.

GEOGRAPHIC DURABILITY

A balanced global manufacturing presence spanning advanced Asian hubs and the expanding $7 billion Arizona campus provides supply chain security. The ramp of the Vietnam facility past breakeven in the first quarter of fiscal 2026 adds cost-efficient operational capacity.

PIPELINE OPTIONALITY

Revenue from 2.5D and High-Density Fan-Out platforms is expected to nearly triple in 2026 to support high-performance computing and AI architectures. Record quarterly revenue across computing and automotive end markets reinforces exposure to secular semiconductor content growth.

The bottom line on risk and reward

Amkor Technology is executing at a high operational standard across its primary advanced packaging lines. The inflection in second-quarter fiscal 2026 gross margin to 16.8%, alongside guidance for approximately $2.0 billion in third-quarter revenue and gross margin breaking 19%, validates the company's technical capabilities in high-density fan-out and communications packaging. Net income growth of 222% year over year in the second quarter highlights the substantial earnings leverage Amkor achieves when advanced packaging lines operate at elevated utilization rates.

However, equity valuation cannot be decoupled from cash flow economics. The company's commitment to deploy $2.5-3.0 billion in capital expenditures during fiscal 2026 represents a historic investment cycle that will absorb cash flow and lift depreciation expenses in subsequent periods. With first-half fiscal 2026 free cash flow already negative at -$307 million, and potential full-year cash burn exceeding $1.5 billion, the near-term risk-reward balance remains unfavorable at current trading levels.

Our probability-weighted valuation framework, assigning 25% weight to our $15.31 Bear target, 50% to our $34.57 Base target, and 25% to our $64.10 Bull target, yields a blended intrinsic equity value of $37.14 per share. This indicates -10.5% downside potential relative to the

reference spot price of $41.50. We view the current market price as fully discounting operational packaging triumphs while understating the duration of capital expenditure headwinds, supporting our Hold recommendation.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

Franchise deep dive

Advanced Packaging Solutions: High-Density Fan-Out and 2.5D Architectures

Amkor's advanced packaging solutions represent the technological core of the enterprise, encompassing cutting-edge High-Density Fan-Out (HDFO), wafer-level chip-scale packaging, and 2.5D multi-die integration platforms. As semiconductor scaling encounters physical interconnect limitations, leading chipmakers increasingly rely on advanced packaging architectures to achieve higher bandwidth, lower latency, and superior power efficiency. Amkor packages these complex semiconductor devices by bridging advanced front-end silicon fabrication with system-level circuit boards, serving as a critical manufacturing partner for computing, communications, and automotive applications.

The operational pivot toward advanced packaging is driving a fundamental mix shift across Amkor's financial statements. In the second quarter of fiscal 2026, advanced products generated $1.557 billion, accounting for 82% of total corporate net sales and growing 27% year over year. This follows the first quarter of fiscal 2026, where advanced packaging contributed $1.37 billion, or 81% of total sales. The multi-die packaging roadmap is increasingly anchored by High-Density Fan-Out and 2.5D platforms, where total revenue is projected to nearly triple throughout fiscal 2026 to satisfy customer demands for artificial intelligence accelerators and high-performance computing processors.

To support this structural demand, Amkor is undertaking an unprecedented capital expenditure program centered on domestic manufacturing infrastructure and HDFO footprint additions. The planned investment for the greenfield campus in Arizona was raised to $7 billion, establishing a cornerstone for the United States semiconductor supply chain. Concurrently, management committed to a fiscal 2026 capital budget of $2.5-3.0 billion, which nearly triples the $905 million deployed in fiscal 2025. This aggressive capacity expansion aims to capture sovereign packaging demand and customer onshoring initiatives, though it inevitably depresses free cash flow during its initial execution phases.

What the recent record shows

- Advanced packaging lines drove record second-quarter fiscal 2026 revenue of $1.898 billion, with advanced products reaching $1.557 billion or 82% of total net sales.
- Platform revenue across 2.5D and HDFO architectures is projected to nearly triple during fiscal 2026, supporting computing and communications demand.

- The Arizona advanced packaging campus investment was raised to $7 billion, reinforcing long-term strategic alignment with domestic semiconductor initiatives.
- Fiscal 2026 capital expenditure was set at an elevated $2.5-3.0 billion envelope, driving first-half free cash flow to -$307 million on $688 million of outlays.

Our read: Amkor's advanced packaging franchise is capturing high-margin demand across artificial intelligence and mobile applications. However, the multi-year capital outlay required to build out the Arizona campus and expand HDFO lines creates a protracted cash flow deficit that limits immediate equity appreciation, keeping our recommendation at Hold.